

LAB ACTIVITY 4

(Due October 7 @23:59)

Please submit the Jupiter notebook file or .py file with your answers. Please don't remove the error messages from the notebook.

1. Write a program with three functions — **main()**, **intro()**, and **ounces()** — to convert cups to fluid ounces. (1 cup = 8 fluid ounces). (4 points)

main() — your main function

intro() — displays an introductory screen with messages for the user

ounces() — accept a number of cups and display the equivalent number of ounces.

2. Define a function **max()** that take three int inputs from the user and print the **largest** number on the screen. (3 points)

3. Define a function **oneMore()** which takes an integer value called **num1** from the user and prints another integer which is one more than the value of **num1**. (4 points)

4. Define a function called **hasRealSolution()** that takes three integer numbers from the user: **num1**, **num2**, and **num3**. If $(num2^2 - 4*num1*num3)$ is negative, the function should print "There is no real solution" otherwise, it "Yes, there is a real solution". (4 points)

5. Define a function named **times_ten()**. The function should accept a float number from the user and display the product of that number multiplied times 10. (3 points)

6. Random number guessing game

Write a program that generates a random number in the range of 1 through 100, and asks the user to guess what the number is. If the user's guess is higher than the random number, the program should display "Too high, try again". If the user's guess is lower than the random number, the program should display "Too low, try again". If the user guesses the number, the application should congratulate the user and generate a new random number so the game can start over. (6 points)